[Only the part which I made is showed. The data used in this project are all mocked-up. Originally written in Thai and then translated into English by the applicant]

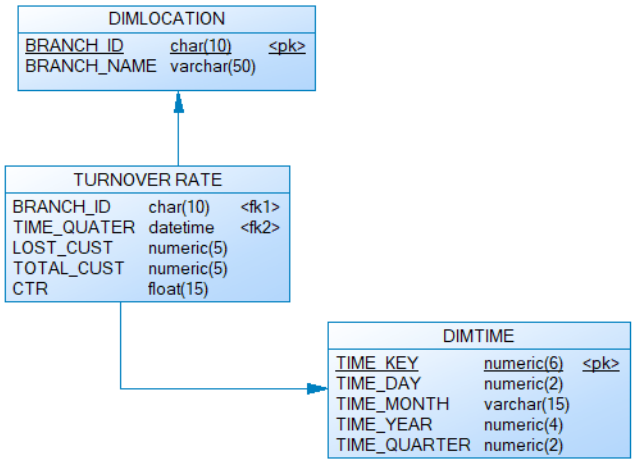
5. How well are we retaining customers?

5.1 KPI: Customer Turnover Rate (CTR)

Formula

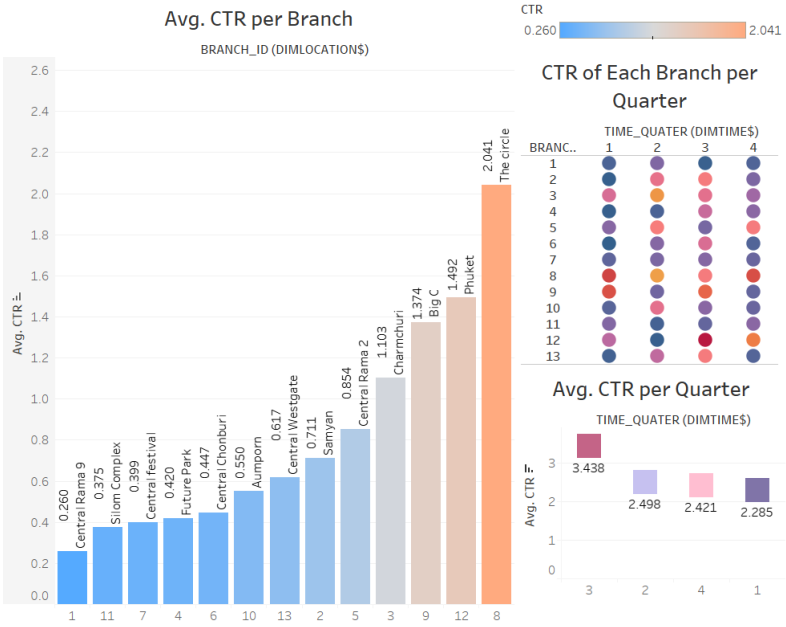
CTR = Lost customers over period *t* / Total number of customers at the end of the period *t*

5.2 Data Warehouse



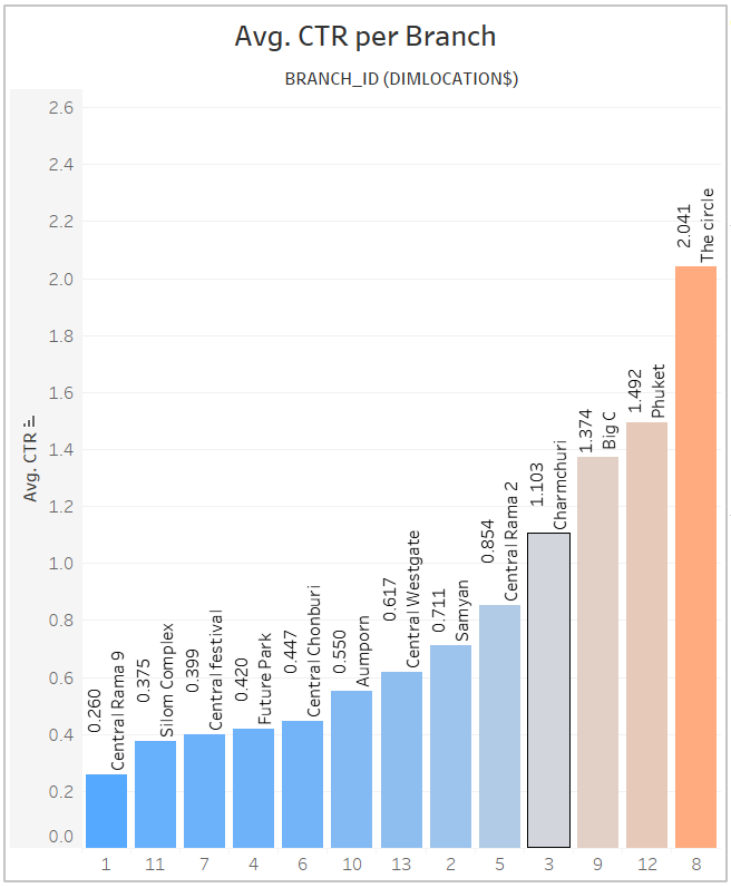
According to the formula of CTR, the data warehouse for this KPI consists of ***Turnover Rate*** table which employs data from two dimensions: ***DIMLOCATION*** and ***DIMTIME.*** *DIMLOCATION* stores data of location of a store, and *DIMTIME* stores data of time when a transaction occurs. Inside the table *Turnover Rate*, there is ***LOST_CUST***, a number of lost customers in a quarter. ***TOTAL_CUST*** represents a number of total customers in the quarter. Finally, ***CTR*** stores the CTR value calculated using the formula given above, and the data stored in *LOST_CUST* and *TOTAL_CUST*.

5.3 Dashboard



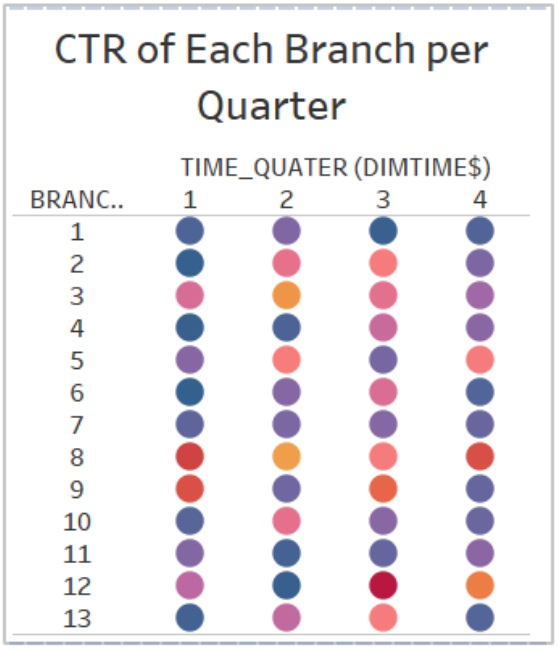
This chart displays a dashboard for the indicator CTR. The dashboard is made of three charts:

Average CTR per Branch



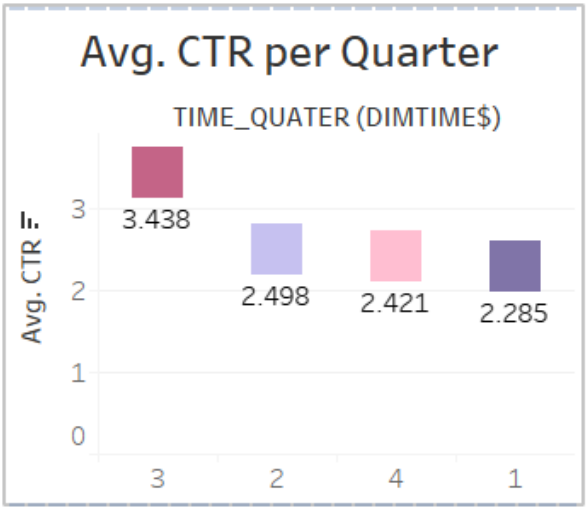
This chart shows the values of average CTR per branch which are calculated from the data collected from year 2019. Each average CTR is sorted ascendingly. Provided by this information, in one hand, it can be claimed that the branch *The circle* lacks the ability to retain customers. The management team should make a policy to mitigate the customer loss. On the other hand, the branch *Central Rama 9* has the smallest value of average CTR. Thus, it can be said that this branch is not at stake and can retain customers.

CTR of Each Branch per Quarter



This chart shows the overall situations of the business by each quarter. Each CTR value is represented by a dot and is differently marked by the tone of the color. The more red a dot gets, the larger a CTR value becomes. Dots marked with reddish tone indicate that the situation of those branches are at high risk of customer loss. Conversely, the more purple a dot gets, the smaller a CTR value becomes: the situations for the branches marked with purplish dots are less concerning. Moreover, the chart also helps us understand that which quarter of each branch should be paid attention to closely.

Average CTR per Quarter



This chart gives us a clear image of average CTR in each quarter. It is obvious that the 3rd quarter has the largest average CTR value. This observation implies that most branches lose their customers at this period of the year. The management team of each branch should come up with a plan or policy to handle the risk of customer loss and consider other factors affecting the risk to maintain their customers.